Acquisition Functions in Bayesian Optimization

Upper Confidence Bound, Expected Improvement, and Thompson Sampling

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1. The Exploration-Exploitation Dilemma

Bayesian optimization addresses the problem of finding the maximum of an unknown function $f(x)$ using as few evaluations as possible. Each evaluation is expensive — in a physical experiment, it might take hours; in a sequential decision process, it consumes a real interaction with the environment. The fundamental tension is between two competing imperatives.

Exploitation says: evaluate where the model's mean $\mu(x)$ is highest, because that is your best current estimate of where the function peaks. If you always exploit, you converge quickly to a local maximum but may never discover a higher peak in an unexplored region.

Exploration says: evaluate where the model's uncertainty $\sigma(x)$ is high, because a surprising discovery there could reveal a much better solution. If you always explore, you learn about the function everywhere but never focus on the promising regions.

The acquisition function is the mathematical mechanism that balances these two imperatives. It maps the GP posterior — specifically, the pair $(\mu(x), \sigma(x))$ — to a single scalar score at every candidate point, and the next evaluation is placed at the point with the highest score. The different acquisition functions we discuss here represent different philosophical choices about how to balance exploration and exploitation.

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2. Upper Confidence Bound (UCB)

2.1 Definition

The UCB acquisition function takes an explicitly optimistic stance. At each candidate point x, it computes:

$$\text{UCB}(x) = \mu(x) + \beta \cdot \sigma(x)$$

where $\beta > 0$ is a scalar that controls the degree of optimism. The idea is to assign each point an **upper confidence bound** on its true function value — a plausible best-case estimate rather than a point estimate. By always choosing the point whose best-case estimate is highest, the algorithm naturally explores uncertain regions (high σ) while also exploiting known good regions (high μ).

2.2 Geometric Interpretation

Think of $\mu(x)$ as the center of a confidence interval at each point, and $\sigma(x)$ as the half-width. UCB always selects the point whose confidence interval extends highest — it is asking: "where might the function be best, given my uncertainty?" As data accumulates, $\sigma(x)$ shrinks in regions that have been evaluated, making those regions look less attractive and naturally driving exploration outward.

2.3 The Role of β

The parameter β controls the width of the confidence intervals used. Small β (close to 0) makes UCB nearly greedy — it selects points where the mean is highest, exploiting heavily. Large β treats uncertainty as highly valuable, exploring aggressively. The theoretical literature (Srinivas et al., 2010) derives a schedule for β_t that grows with t at a rate depending on the input space dimension and kernel complexity. Under this schedule, the **cumulative regret** — the total loss compared to always choosing the optimal point — grows sublinearly, meaning the algorithm converges to the optimum. Using a fixed β loses this guarantee but works well in practice for short horizons.

2.4 UCB in Multi-Armed Bandits

In the discrete multi-armed bandit setting (where arms are distinct options rather than points in a continuous space), the standard UCB formula is:

$$\text{UCB}_t(a) = \hat{\mu}_t(a) + \sqrt{\frac{2\ln t}{N_t(a)}}$$

where $\hat{\mu}_t(a)$ is the empirical average reward for arm a, t is the total number of pulls, and $N_t(a)$ is the number of times arm a has been pulled. The exploration bonus $\sqrt{2\ln t / N_t(a)}$ grows for rarely-pulled arms and shrinks as more data accumulates — a natural count-based measure of uncertainty that requires no kernel or GP machinery.

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3. Expected Improvement (EI)

3.1 Motivation

UCB uses the full standard deviation $\sigma(x)$ regardless of whether the mean $\mu(x)$ is above or below the current best. This means UCB will enthusiastically explore a region with high uncertainty even if the mean is far below the current best — there's very little realistic chance of improvement there, yet UCB is drawn in by the uncertainty. Expected Improvement corrects this by being more surgical: it asks precisely how much improvement you can realistically expect, accounting for the shape of the posterior distribution at x.

3.2 Derivation from First Principles

Define the improvement at point x as the amount by which evaluating x beats the current best $f^* = \max_i y_i$:

$$I(x) = \max(0,\ f(x) - f^*)$$

The asymmetry — $\max(0, \cdot)$ rather than just $f(x) - f^*$ — reflects the fact that if the evaluation turns out worse than f^*, you simply discard it. You only benefit from improvements. Expected Improvement is the expectation of this quantity over the GP posterior:

$$\text{EI}(x) = \mathbb{E}[\max(0,\ f(x) - f^*)]$$

Since the GP tells us $f(x) \sim \mathcal{N}(\mu, \sigma^2)$, we can write this expectation as an integral. The $\max(0, \cdot)$ term kills the integrand everywhere $f(x) < f^*$, so:

$$\text{EI}(x) = \int_{f^*}^{\infty} (y - f^*) \cdot \frac{1}{\sigma}\phi\!\left(\frac{y - \mu}{\sigma}\right) dy$$

where ϕ is the standard normal density.

3.3 Closed-Form Solution

The substitution $z = (y - \mu)/\sigma$ transforms the integral into:

$$\text{EI}(x) = \int_{z^*}^{\infty} (\mu + \sigma z - f^*)\,\phi(z)\,dz$$

where $z^* = (f^* - \mu)/\sigma$. Splitting into two integrals and defining $Z = (\mu - f^*)/\sigma = -z^*$:

$$\text{EI}(x) = (\mu - f^*)\underbrace{\int_{-Z}^{\infty}\phi(z)\,dz}_{=\Phi(Z)} + \sigma\underbrace{\int_{-Z}^{\infty}z\,\phi(z)\,dz}_{=\phi(Z)}$$

The second integral uses the identity $\int_{z^*}^{\infty} z\,\phi(z)\,dz = \phi(z^*)$, which follows from $z\,\phi(z) = -\phi'(z)$ and the fundamental theorem of calculus. The final closed form is:

$$\boxed{\text{EI}(x) = (\mu - f^*)\,\Phi(Z) + \sigma\,\phi(Z), \quad Z = \frac{\mu - f^*}{\sigma}}$$

where Φ is the standard normal CDF and ϕ is the standard normal PDF.

3.4 Geometric Interpretation of the Two Terms

The two terms in the EI formula capture the two contributions to expected improvement with beautiful clarity.

The **exploitation term** $(\mu - f^*)\,\Phi(Z)$ represents the improvement you would get if the sampled function value equaled the posterior mean, weighted by the probability that the sample actually exceeds f^*. When $\mu \gg f^*$, the mean prediction is well above the current best, $\Phi(Z) \approx 1$, and nearly the full expected gain is realized. When $\mu \ll f^*$, $\Phi(Z) \approx 0$ and this term vanishes — correctly, because the posterior mean is too low to realistically beat f^*.

The **exploration term** $\sigma\,\phi(Z)$ is maximized when $Z = 0$, i.e., when $\mu = f^*$ — exactly on the boundary of improvement. This is where uncertainty contributes most to expected improvement: you are poised at the decision boundary, and any positive deviation from the mean would yield a gain. The Gaussian PDF $\phi(Z)$ acts as a concentration function, ensuring this exploration value is highest near the boundary and falls away symmetrically on either side.

3.5 EI Collapse and the Sliding Window Best

A subtle failure mode arises in non-stationary environments. If the environment shifts and rewards degrade, the current best f^* may become outdated — it reflects a past environment that no longer exists. With f^* too high relative to the current mean, Z becomes a large negative number, both $\Phi(Z)$ and $\phi(Z)$ collapse toward zero, and EI becomes negligibly small everywhere. The acquisition function loses all useful signal.

The most practical fix is to compute f^* over a **sliding window** of recent observations rather than the global historical maximum:

$$f^*_t = \max_{i \in [t-W, t]} y_i$$

where W is the window size. This allows f^* to decay when rewards degrade, preventing EI from collapsing against an outdated benchmark. The window size W is a hyperparameter controlling how quickly the system forgets past performance — a genuine tradeoff between stability and adaptivity.

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4. Thompson Sampling

4.1 The Key Idea

Thompson Sampling takes a philosophically different approach to the exploration-exploitation tradeoff. Rather than computing a deterministic acquisition value from the posterior statistics (μ, σ), it directly samples a **random function** from the GP posterior and acts as if that sample were the true function.

The procedure is: draw $\tilde{f} \sim \text{GP posterior}$, then select $x^* = \arg\max_x \tilde{f}(x)$.

The exploration-exploitation balance emerges automatically from the randomness of the sample. Where the GP is confident (small σ), all samples look similar — they peak in roughly the same place, so the algorithm consistently exploits. Where the GP is uncertain (large σ), samples vary wildly — different samples peak in different places, so across multiple iterations the algorithm explores broadly.

4.2 Sampling from a Gaussian Posterior

To draw a sample from the GP posterior at a set of G candidate points, we use the **Cholesky sampling method**. The posterior at those points is a multivariate Gaussian $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where $\boldsymbol{\mu}$ is the G-dimensional mean vector and Σ is the $G \times G$ covariance matrix.

A sample is drawn as:

$$\tilde{f} = \boldsymbol{\mu} + L\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$$

where $L = \text{chol}(\Sigma)$. This works because if $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$, then $L\mathbf{z} \sim \mathcal{N}(\mathbf{0}, LL^T) = \mathcal{N}(\mathbf{0}, \Sigma)$, so $\boldsymbol{\mu} + L\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

4.3 Thompson Sampling vs UCB and EI

Thompson Sampling has several properties that distinguish it from UCB and EI.

No tuning parameter. UCB requires choosing β, and EI is sensitive to the window size for f^*. Thompson Sampling requires no such choices — the exploration-exploitation balance is fully determined by the posterior.

Robustness to EI collapse. Because Thompson Sampling uses the full posterior distribution rather than comparing to f^*, it is immune to the collapse problem that afflicts EI in non-stationary environments. Even when the posterior mean is uniformly low, different samples will peak in different places, ensuring continued exploration.

Natural diversity in parallel settings. When selecting multiple evaluation points simultaneously (batch Bayesian optimization), drawing multiple independent Thompson samples automatically produces diverse candidates — each sample reflects a different plausible world. UCB and EI, being deterministic, would recommend the same point repeatedly.

Theoretical properties. Thompson Sampling satisfies a Bayesian regret bound: the expected cumulative regret over T rounds is bounded by $\tilde{O}(\sqrt{T})$ under mild conditions, matching the best known bounds for UCB.

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5. The Softmax Policy for Discrete Arm Selection

When choosing among a discrete set of arms (rather than optimizing over a continuous space), a useful policy is the **softmax** (or Boltzmann) distribution over arm scores s_a:

$$\pi(a) = \frac{\exp(s_a / \tau)}{\sum_{a’} \exp(s_{a’} / \tau)}$$

The temperature parameter $\tau > 0$ controls the concentration of the distribution. As $\tau \to 0$, the policy concentrates entirely on the arm with the highest score (greedy). As $\tau \to \infty$, the policy becomes uniform over all arms (pure exploration). At intermediate τ, the policy is smooth and differentiable with respect to the scores, which enables gradient-based optimization of policies in reinforcement learning.

In practice, to avoid numerical overflow when scores differ by large amounts, the **log-sum-exp trick** is used: subtract the maximum score before exponentiating.

$$\pi(a) = \frac{\exp((s_a - s_{\max}) / \tau)}{\sum_{a’} \exp((s_{a’} - s_{\max}) / \tau)}$$

This is numerically equivalent but prevents any individual exponential from overflowing.

See also: Gaussian Process Regression; Multi-Output GP and Coregionalization; The Two-Timescale Bayesian Update